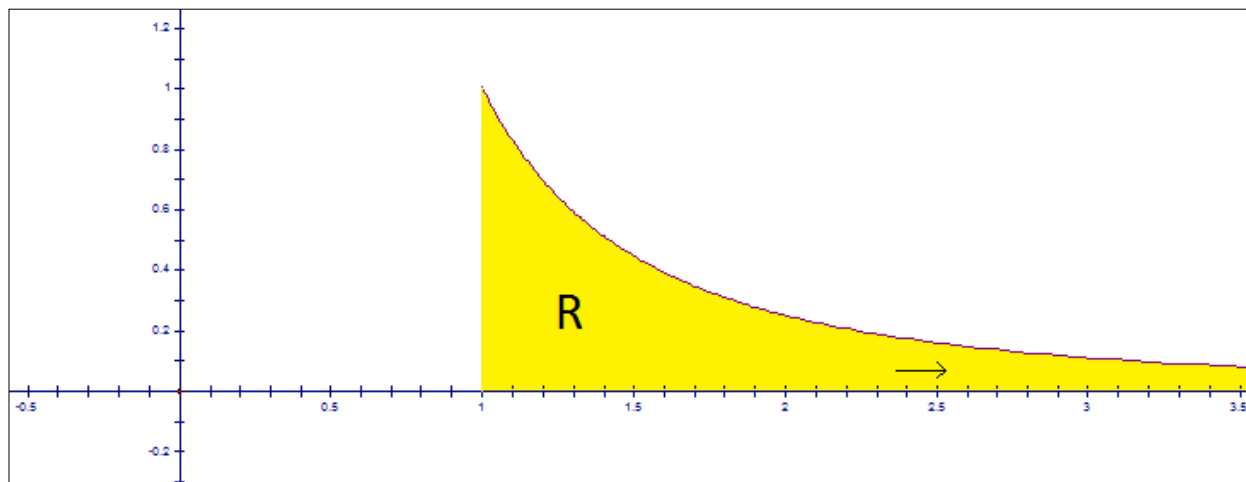


2.7 Improper Integrals

Motivation:

Suppose we want to find the area of the region, R , given by $R = \left\{ (x, y) \mid x \geq 1 \text{ and } 0 \leq y \leq \frac{1}{x^2} \right\}$. A picture of this region is shown below (in yellow):



The key question here is: does the area under the above curve approach a finite number?

To answer this question we can use integration. Specifically we calculate:

$$\lim_{t \rightarrow \infty} \left(\int_1^t \frac{1}{x^2} dx \right) = \lim_{t \rightarrow \infty} \left(\left. \frac{-1}{x} \right|_1^t \right) = \lim_{t \rightarrow \infty} \left(\frac{-1}{t} + 1 \right) = 1$$

So, we say that the area of the region R is one.

Definition (Improper Integral of Type 1):

1. If $\int_a^t f(x) dx$ exists for every number $t \geq a$, then $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$ provided this limit exists (as a finite number).
2. If $\int_t^b f(x) dx$ exists for every number $t \leq b$, then $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$ provided this limit exists (as a finite number).

Definition: The improper integrals $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called **convergent** if the corresponding limit exists (as a finite number), and are called **divergent** if the limit does not exist.

Definition: If both $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{\infty} f(x)dx$ where a can be taken to be any real number.

Note: To shed some light as to why a can be taken to be any real number in this definition you can consider the following question:

If $\int_{-\infty}^{\infty} f(x)dx$ is convergent and a and b are real numbers, show that:

$$\int_{-\infty}^a f(x)dx + \int_a^{\infty} f(x)dx = \int_{-\infty}^b f(x)dx + \int_b^{\infty} f(x)dx$$

Example 1 (you try): Determine whether $\int_4^{\infty} \frac{1}{\sqrt{x}} dx$ is convergent or divergent. If it is convergent evaluate the integral.

We see that $\int_4^{\infty} \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow \infty} \int_4^t \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow \infty} \left(2\sqrt{x} \Big|_4^t \right) = \lim_{t \rightarrow \infty} (2\sqrt{t} - 4) = \infty$. Since the limit does not exist as a finite number we say that the improper integral is divergent.

Example 2: Evaluate $\int_{-1}^{\infty} xe^{-x} dx$

We use tabular integration (or integration by parts) and find:

$$\begin{aligned} \int_{-1}^{\infty} xe^{-x} dx &= \lim_{t \rightarrow \infty} \int_{-1}^t xe^{-x} dx = \lim_{t \rightarrow \infty} \left(\left[-xe^{-x} - e^{-x} \right]_{-1}^t \right) \\ &= \lim_{t \rightarrow \infty} (-te^{-t} - e^{-t} - (e - e)) = \lim_{t \rightarrow \infty} (-te^{-t} - e^{-t}) . \end{aligned}$$

Now, we use l'Hospital's Rule and we see: $\lim_{t \rightarrow \infty} (-te^{-t} - e^{-t}) = \lim_{t \rightarrow \infty} \left(\frac{-t}{e^t} \right) - \lim_{t \rightarrow \infty} \left(\frac{1}{e^t} \right) = \lim_{t \rightarrow \infty} \left(\frac{-1}{e^t} \right) - 0 = 0$.

Example 3: Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Using the above definition, we take $a = 0$ (since it will be convenient). By definition we know that

$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx$ provided that the two integrals on the right hand side of this equation are convergent. Now, we calculate:

$$\begin{aligned} \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx + \lim_{s \rightarrow \infty} \int_0^s \frac{1}{1+x^2} dx \\ &= \lim_{t \rightarrow -\infty} \left(\tan^{-1}(x) \Big|_t^0 \right) + \lim_{s \rightarrow \infty} \left(\tan^{-1}(x) \Big|_0^s \right) \\ &= \lim_{t \rightarrow -\infty} \left(\tan^{-1}(0) - \tan^{-1}(t) \right) + \lim_{s \rightarrow \infty} \left(\tan^{-1}(s) - \tan^{-1}(0) \right) \\ &= \frac{\pi}{2} + \frac{\pi}{2} = \pi. \end{aligned}$$

The next definition gives us a way of dealing with vertical asymptotes as they arise in functions that we integrate.

Definition (Improper integral of Type 2):

1. If f is continuous on $[a, b)$ and discontinuous at b , then $\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$ provided this limit exists (as a finite number).
2. If f is continuous on $(a, b]$ and discontinuous at a , then $\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$ provided this limit exists (as a finite number).

Note: The notions of convergent and divergent are defined similar to the way in which we defined them for Improper Integrals of Type 1.

Definition: If f has a discontinuity at c , where $a < c < b$ (and f is continuous on $[a, c)$ and $(c, b]$) and both $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ are convergent, then we define $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

Example 4: Evaluate $\int_0^{\frac{\pi}{2}} \tan(x) dx$ if possible.

We first note that $g(x) = \tan(x)$ is continuous on $\left[0, \frac{\pi}{2}\right)$ and discontinuous at $\frac{\pi}{2}$. So, by the above definition we have:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \tan(x) dx &= \lim_{t \rightarrow \frac{\pi}{2}^-} \int_0^t \tan(x) dx = \lim_{t \rightarrow \frac{\pi}{2}^-} \left(-\ln(\cos(x)) \Big|_0^t \right) \\ &= \lim_{t \rightarrow \frac{\pi}{2}^-} \left(-\ln(\cos(t)) + \ln(\cos(0)) \right) = \infty .\end{aligned}$$

Thus, the given improper integral is divergent.

Example 5 (you try): Consider the following problem: Evaluate $\int_0^3 \frac{1}{x-1} dx$ if possible. Explain what is wrong

with the following solution: $\int_0^3 \frac{1}{x-1} dx = \ln|x-1| \Big|_0^3 = \ln(2) - \ln(1) = \ln(2)$.

This solution fails to take into account the fact that $h(x) = \frac{1}{x-1}$ has a discontinuity at 1. Therefore, it is an improper integral and must be computed in terms of limits.

Note: It can be shown that the above integral is actually divergent.

Question:

Does $\lim_{t \rightarrow \infty} \int_0^t e^{-x^2} dx$ converge to a finite number?

This question is particularly difficult because we know that we cannot find an antiderivative for $f(x) = e^{-x^2}$ in terms of elementary functions. The next theorem will give us a tool which can be helpful in answering questions like these.

Theorem (Comparison Theorem):

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

1. If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent.
2. If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.

Note: This theorem is called the comparison theorem by the book. Since the book omits the proof of this theorem, I too will omit it. However, if you would like to discuss the proof of this theorem feel free to stop by office hours.

Example 6: Prove that $\int_0^\infty e^{-x^2} dx$ is convergent.

We know that $\int_0^\infty e^{-x^2} dx = \lim_{t \rightarrow \infty} \int_0^t e^{-x^2} dx$. Now, by the properties of definite integrals we have:

$$\int_0^\infty e^{-x^2} dx = \lim_{t \rightarrow \infty} \int_0^t e^{-x^2} dx = \lim_{t \rightarrow \infty} \left(\int_0^1 e^{-x^2} dx + \int_1^t e^{-x^2} dx \right) = \int_0^1 e^{-x^2} dx + \lim_{t \rightarrow \infty} \int_1^t e^{-x^2} dx.$$

We note that $\int_0^1 e^{-x^2} dx$ is just an ordinary definite integral, and hence is finite. Now, we note that

$x \geq 1 \Rightarrow x^2 \geq x \Rightarrow -x^2 \leq -x$. This implies that $0 \leq e^{-x^2} \leq e^{-x}$ on the interval $[1, \infty)$.

This is a very nice relation to have since $\int_1^\infty e^{-x} dx$ is easy to evaluate. Specifically, we calculate:

$$\int_1^\infty e^{-x} dx = \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx = \lim_{t \rightarrow \infty} \left(-e^{-x} \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(-e^{-t} + e^{-1} \right) = \frac{1}{e}.$$

By part 1 of the comparison theorem, we know that

this means $\lim_{t \rightarrow \infty} \int_1^t e^{-x^2} dx$ is convergent. This then implies that $\int_0^\infty e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \lim_{t \rightarrow \infty} \int_1^t e^{-x^2} dx$ is convergent.

■

Note: Usually the most challenging part of a proof that uses the comparison theorem is finding a “nice” integral to which to compare the integral of interest.

Example 7: Gabriel's horn is the solid obtained by rotating the region $R = \left\{ (x, y) \mid x \geq 1 \text{ and } 0 \leq y \leq \frac{1}{x} \right\}$ about the x -axis, find whether or not the surface area is finite.

We are interested in surface area of the solid obtained by revolving the curve given by $y = \frac{1}{x}$ where $x \geq 1$ about the x -axis. By definition the desired surface area is:

$$\begin{aligned} 2\pi \left(\lim_{t \rightarrow \infty} \int_1^t y \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \right) &= 2\pi \left(\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} \sqrt{1 + \left(\frac{-1}{x^2} \right)^2} dx \right) = 2\pi \left(\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx \right) \\ &= 2\pi \left(\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} \sqrt{\frac{x^4 + 1}{x^4}} dx \right) = 2\pi \left(\lim_{t \rightarrow \infty} \int_1^t \frac{\sqrt{x^4 + 1}}{x^3} dx \right). \end{aligned}$$

We note that for $x \geq 1$ we have $\frac{\sqrt{x^4 + 1}}{x^3} \geq \frac{\sqrt{x^4}}{x^3} = \frac{1}{x} \geq 0$. Also, we

have that $\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} (\ln |x|_1^t) = \infty$. So, by the comparison

theorem: $\lim_{t \rightarrow \infty} \int_1^t \frac{\sqrt{x^4 + 1}}{x^3} dx$ is divergent and we know that the

surface area of Gabriel's horn is infinite.

Note: Interestingly we can also prove that the volume of Gabriel's horn is finite!

Volume Calculation:

$$V = \int_1^{\infty} \pi \left(\frac{1}{x} \right)^2 dx = \lim_{t \rightarrow \infty} \int_1^t \pi \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\pi \frac{1}{x} \right]_1^t = \lim_{t \rightarrow \infty} \left[-\frac{\pi}{t} + \frac{\pi}{1} \right] = 0 + \pi = \pi \quad \text{A finite volume!!!}$$

So Gabriel's Horn has:

- Infinite surface area &
- Finite volume

This is like a paint can that cannot hold enough paint to paint its own exterior!

